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Introduction

I am a final-year Ph.D. candidate at CUHK (expected June 2026), advised by **Prof. Chi-Wing FU**. My research focuses on robotic manipulation and embodied skill modeling, in collaboration with **Prof. Yun-Hui Liu** (CUHK T-Stone Robotics Institute and Hong Kong Centre for Logistics Robotics) and **Prof. Mingyu Ding** (UNC-Chapel Hill). My long-term goal is to develop unified frameworks that enable robots to acquire, evaluate, and generalize human manipulation skills.

Education

The Chinese University of Hong Kong
Ph.D. in Computer Science

(Aug. 2021 - Jun. 2026)

Supervisor: **Prof. Chi-Wing Fu**

Sun Yat-sen University
M.Phil. in Computer Science

(Aug. 2018 - Jul. 2021)

Supervisor: **Prof. Wei-Shi Zheng**

Sun Yat-sen University
B.Sc. in Computer Science

(Aug. 2014 - Jun. 2018)

GPA: 3.8 out of 4.0

Publications

First-Author Works

- * **Jia-Hui Pan**, Yeok Tatt Cheah, Zhengzhe Liu, Ka-Hei Hui, Xiaojie Gao, Pheng-Ann Heng, Yun-Hui Liu, and Chi-Wing Fu. "OPA-Pack: Object-Property-Aware Robotic Bin Packing." *Conditionally accepted to IEEE Transactions on Robotics (T-RO)*.
- * **Jia-Hui Pan**, Zhengzhe Liu, Ka-Hei Hui, Kai Chen, Xiaojie Gao, Pheng-Ann Heng, Qi Dou, Yun-Hui Liu, Mingyu Ding, and Chi-Wing Fu. "Human-Intuition-driven Tree Packing for Diverse Everyday Objects." *Under review in IEEE/ASME Transactions on Mechatronics*.
- * **Jia-Hui Pan**, Xiaojie Gao, Ka-Hei Hui, Shize Zhu, Yun-Hui Liu, Pheng-Ann Heng, and Chi-Wing Fu. "PPN-Pack: Placement Proposal Network for Efficient Robotic Bin Packing." *IEEE Robotics and Automation Letters (RA-L)*, 2024.
- * **Jia-Hui Pan**, Ka-Hei Hui, Xiaojie Gao, Shize Zhu, Yun-Hui Liu, Pheng-Ann Heng, and Chi-Wing Fu. "SDF-Pack: Towards Compact Bin Packing with Signed-Distance-Field Minimization." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- * **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Adaptive Action Assessment." *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2021.

- * **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Action Assessment by Joint Relation Graphs." *IEEE International Conference on Computer Vision (ICCV)*, 2019. (Oral)

Selected Collaborative Works

- Weiliang Tang, Dong Jing, **Jia-Hui Pan**, Zhiwu Lu, Yunhui Liu, Li Li, Mingyu Ding, and Chi-Wing Fu. "Incentivizing Multimodal Reasoning in Large Models for Direct Robot Manipulation." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- Weiliang Tang, **Jia-Hui Pan**, Yunhui Liu, Masayoshi Tomizuka, Li Li, Chi-Wing Fu, and Mingyu Ding. "Geometric Constraints as General Interfaces for Robot Manipulation" *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- Kai Chen, Chengkun Li, Chang Tu, **Jia-Hui Pan**, Yiyao Ma, Wei Chen, Zhongxiang Zhou, Xuecheng Xu, Stephen James, Chi-Wing Fu, Rong Xiong, Pieter Abbeel, Yun-Hui Liu, and Qi Dou. "A Retrieval-Augmented Framework Enabling VLM Spatial Awareness for Object-Centric Robot Manipulation." *Science Robotics*, 2026.
- Rui Cao, **Jia-Hui Pan**, Yichuan Li, Chi-Wing Fu, Jiangliu Wang, and Yun-Hui Liu. "MMGNet: Multi-Modal 6-DoF Grasp Synthesis for Cluttered Scenes." *Under review in IEEE/ASME Transactions on Mechatronics*.
- Rui Cao, **Jia-Hui Pan**, Gang Wang, Yin Zhang, Qiwei Meng, Runsong Zhu, Wei Chen, Chi-Wing Fu, Jiangliu Wang, and Yun-Hui Liu. "Uncertainty-aware Metric Depth Learning for Material-Agnostic Object Perception" *Under review in IEEE Transactions on Robotics*.
- Weiliang Tang, Jialin Gao, **Jia-Hui Pan**, Gang Wang, Li Erran Li, Yun-Hui Liu, Mingyu Ding, Pheng-Ann Heng, and Chi-Wing Fu. "Rethinking Intermediate Representation for VLM-Based Robot Manipulation." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- Weiliang Tang, **Jia-Hui Pan**, Wei Zhan, Jianshu Zhou, Huaxiu Yao, Yun-Hui Liu, Masayoshi Tomizuka, Mingyu Ding, and Chi-Wing Fu. "Embodiment-Agnostic Action Planning via Object-Part Scene Flow." *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- Jibin Gao, **Jia-Hui Pan**, Shao-Jie Zhang, and Wei-Shi Zheng. "Automatic Modeling for Interactive Action Assessment." *International Journal of Computer Vision (IJCV)*, 2023.
- Shao-Jie Zhang, **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Adaptive Stage-Aware Assessment Skill Transfer for Skill Determination." *IEEE Transactions on Multimedia (TMM)*, 2023.
- Shao-Jie Zhang, **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Semi-Supervised Action Quality Assessment with Self-Supervised Segment Feature Recovery." *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 2022.
- Jibin Gao, Wei-Shi Zheng, **Jia-Hui Pan**, Chengying Gao, Yaowei Wang, Wei Zeng, and Jianhuang Lai. "An Asymmetric Modeling for Action Assessment." *European Conference on Computer Vision (ECCV)*, 2020.